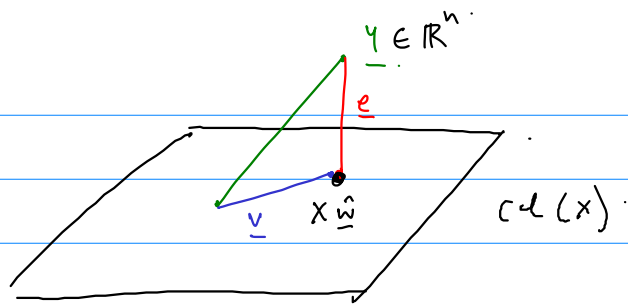




$X: n \times p$ (tall).

$\underline{e} = \underline{y} - X\underline{\hat{w}}$ is perpendicular to every col of X .

$$X^T \underline{e} = \underline{0} \quad \therefore \underline{e} \in \text{null}(X^T)$$

* Any vector $\underline{y} \in \mathbb{R}^n = \underbrace{\underline{v}}_{\text{vector in col}(X) = \text{row}(X^T)} + \underbrace{\underline{e}}_{\text{vector in null}(X^T)}$

$A: m \times n$

* Every vector in $\text{row}(A)$ is $\perp$ to every vector in $\text{null}(A)$.

By definition if $\underline{x} \in \text{null}(A)$

$$A\underline{x} = \underline{0} \quad \begin{pmatrix} \text{---} r_1^T \text{---} \\ \text{---} r_m^T \text{---} \end{pmatrix} \begin{pmatrix} 1 \\ \underline{x} \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$\therefore$ Every row of A , and hence all linear comb of rows of A are $\perp$ to $\underline{x}$

* Any vector $\underline{y} \in \mathbb{R}^n$ can be written as $\underline{v} + \underline{e}$ where $\underline{v} \in \text{row}(A)$ $\underline{e} \in \text{null}(A)$

$$\text{row}(A) \cap \text{null}(A) = \{\underline{0}\}$$

$$\begin{aligned} \text{Suppose } \underline{z} \in \text{row}(A). \quad \therefore \underline{z}^T &= \underline{w}^T A. \quad \text{If } \underline{z} \in \text{null}(A), \\ \underline{z}^T \underline{z} &= (\underline{w}^T A) \underline{z} = \underline{w}^T (A \underline{z}) = \underline{w}^T \underline{0} = 0. \\ &\quad \uparrow \\ &\quad \underline{z} \in \text{null}(A) \end{aligned}$$

Basis for $\mathbb{R}^n$. (Suppose A has k pivots, A^T has k pivots)

union { Basis for $\text{row}(A)$: Elimination on A^T , pick pivot cols of A^T , k vectors.

{ Basis for $\text{null}(A)$: n vectors.

$\therefore$ Basis for $\mathbb{R}^n$ has $k+n$ vectors.

$$\therefore k+n=n \quad \text{or} \quad n=n-k.$$

$$A \underline{x} = \lambda \underline{x}$$

$$\frac{d}{dt} e^{zt} = z e^{zt} \quad z \text{ is a complex number.}$$

$$X = \begin{bmatrix} -\underline{x}_1^T \\ \vdots \\ -\underline{x}_n^T \end{bmatrix}$$

